

TRANSPORT PHENOMENA

ChE 319 – Spring 2026

Unique # – 16470

Meets in CPE 2.220 on MWF from 1:00 pm – 1:50 pm

Instructor - Prof. C. B. Mullins, CPE 4.418, mullins@che.utexas.edu

TEXTBOOK:

R. B. Bird, W. E. Stewart and E. N. Lightfoot, *Transport Phenomena*, 1st Ed., John Wiley & Sons, Inc., 1960. [*Transport Phenomena*, 1st Ed.](#) [The second edition will also work but it is much more expensive] You may or may not find the book useful but I will be assigning some reading from it. You are not required to purchase the book.

PREREQUISITES:

ChE 317-Introduction to Chemical Engineering Analysis and Differential Equations.

RECITATION SESSIONS:

W 4:00 – 6:00 p.m. CPE 2.220 (all recitations here)

OFFICE/HELP SESSIONS:

Saturdays or Sundays TBD
Mondays from 5 – 7 pm in CPE 2.216
Tuesdays from 5 – 7 pm in CPE 2.216

MATERIALS:

Classroom lecture materials and homework assignments will be posted on our Canvas web-site in a timely fashion.

CELL PHONES/LAPTOPS:

Please don't use cell phones during class. Please turn them off and stow them away before class begins. Points will be deducted from your final average for using electronic devices during class. Laptops and tablets can be used for taking notes but please don't do email, messaging, surfing the web, etc. during this time If you are going to come to class, engage ... it is absolutely the best use of your time (keep in mind how much it is costing you and your family).

Although these electronic devices can be quite useful they can also be very distracting to you, other students and me. This class is quite difficult and you'll need to be fully attentive to be successful and I want all of you to be highly, highly successful. Your cooperation is greatly appreciated!!

LECTURES:

It is important and expected that you will attend all lectures. I will try to do my part to make the lectures worth attending. Please be prompt since late arrivals tend to distract others in the class. There is a strong correlation between class attendance and participation in course activities and understanding/performance in the course, therefore, I strongly advise you to attend lectures. **If you miss class or arrive late and miss receiving a handout please get the materials you missed from one of your classmates. Further, missed lecture quizzes (see section ahead) cannot be made up.**

Please be courteous to others during class: I typically don't want you speaking with each other during class. Thank you in advance!!

There are several important concepts and techniques in this course that I will not be able to discuss in great detail in lecture. However, I plan to assign homework problems and generate test questions that will deal with some or all of these concepts. You will be expected to learn about these

techniques outside of class from reading the textbook or library research. This is good practice!

Further, study of the appropriate sections of the text prior to lecture enhances the learning experience for me as well as you since our interaction can be much more of a two-way dialogue rather than the typical "I speak, you listen" discourse. Such advanced preparation also allows us to rapidly cover the rudiments so that we may address the finer points of the subject more thoroughly.

I can assign $\pm$ points to your final average based on your professionalism and participation. For example, if you are *consistently* wide-awake in class asking good questions during every class period, answering the questions I pose to the class, and arriving in class ready to go before the bell rings, I'll add points to your final average at the end of the semester (up to 3 points). Additionally, if in making up sample problems/solutions for student colleagues while preparing for exams you create a very good problem/solution that you share with me, this action could contribute to additional points. Alternatively, if you use electronic devices in our class, arrive late, and/or cause a disturbance in class I will deduct points from your final average please give me a really positive impression of you.

HOW TO LEARN:

A book for lay people entitled *Make it Stick* by scholars Peter C. Brown, Henry L. Roediger, III, and Mark A. McDaniel, suggests the following for optimal learning:

(i) learning is more durable and lasting when we put effort in to it (after all, the brain needs to physically change for us to learn new things),

(ii) re-reading text and cramming are ineffective (these are typically low activity (or passive) practices),

(iii) recalling information (or simple self-testing) is more effective than re-reading (self-testing is very active learning). For example, (a) try to summarize or outline what you have read and/or (b) work example problems from books or lecture without assistance, (c) work (or re-work) a homework problem without using your notes or the book.

(iv) several "short" intense study sessions spread out over an extended period (many days or weeks) is more effective than one "long" study session (I recommend turning off electronic devices and other distractions), and

(v) trying (really hard) to solve a problem *prior* to being taught the solution leads to much better learning (again, this is very active learning and what you will be expected to do once you are on the job in industry).

I have found the book *Make it Stick* to be quite interesting and useful and some portions of the book are available for free on <http://books.google.com/>.

You may also wish to check out the *free* Coursera online course entitled "Learning How to Learn": <https://www.coursera.org/learn/learning-how-to-learn>. In this course the so-called "Pomodoro Technique" is described and this may be helpful to you.



Here is how the pomodoro technique works:

1. Choose the task you will work on from your priority list.
2. Set your timer (the pomodoro – tomato in Italian) for 25 minutes. Turn off your cell phone (and eliminate other distractions).
3. *Immerse* yourself in the task for 25 minutes. If you think of something you need/want to do in the future make a quick note of it so that you can remember to address this at the end of 25 minutes.
4. When the pomodoro buzzes feel free to take a break (or continue with work if that feels right).
5. Start over after your break and you are refreshed.

LECTURE QUIZZES:

We will have a short quiz at the beginning of nearly every lecture period beginning with the 4th lecture period on Wednesday Jan. 21st. On occasion, just to keep you on your toes, I will also give you a short quiz at random times during random lecture periods (this will contribute to active learning). I will give some quizzes near the end of lecture periods over material covered during the lecture to motivate lecture attentiveness. These quizzes will typically only require a small calculation or relatively simple explanation. **You will receive credit (full) only if your answer is perfect.** If there is an error in your answer you will receive a zero for that particular quiz ... I really want you to be careful and thoughtful here. In order to be fair to everyone I also need you to hand these quizzes to me in a timely manner ... if 3 minutes is allowed for completing a quiz then your quiz must be handed in at the end of three minutes and not a second later. I need to strictly enforce this so that everyone is treated fairly. I will typically tell you when one minute is remaining for completing the quiz. There will be a minimum of 10 of these quizzes throughout the semester and probably no more than 20 quizzes. **Since there are so many lecture quizzes and each one of them is not worth all that much, missed quizzes cannot be made up and will be scored as a zero. However, I will drop your three lowest lecture quiz grades.** Lecture quizzes will not be returned to you. The results from lecture quizzes will represent 10% of your final grade. **Bring paper to class each day for the lecture quiz.**

HOMEWORK POLICY:

There is a homework assignment for nearly every week during the term. In addition to the traditional mathematically-oriented problems I will also have some readings for you regarding “soft-skills” which is discussed next. Homework will not be collected, however, you will have a closed book/closed notes quiz over the homework set and readings that will be administered during the recitation. These recitation quizzes will be composed of problems very similar to homework problems and/or verbatim homework problems in addition to relevant examples from the text and some questions from the “soft-skills” readings that are provided. The recitation homework quiz grades will contribute 15% to your final grade. **Recitation quizzes cannot be made up, however your lowest recitation HW quiz score will be dropped.** If more than one recitation quiz is missed the additional missed quizzes will be scored as a zero. I expect you all to score perfectly or nearly perfectly on recitation quizzes (since you will have had the problem set in your possession for a week or longer in addition to the assistance of me, the TA, and the tutor(s) for troublesome problems). I have instructed the grader to grade the quizzes with this in mind (i.e., if it is clear to the grader when they grade your quiz

that you are clueless about how to work a problem you will receive zero credit).

You may work together with your classmates on homework but don't deny yourself the opportunity to "struggle" with difficult concepts and problems (by having friends "spoon-feed" you – again, think “active” learning). It is my opinion that one learns a great deal in this endeavor homework problems are your means of "practicing" engineering just as an athlete practices to improve his/her athletic ability. If I can help you meet other people in the class to form a study group please let me know.

Many of the assigned homework problems will be very challenging indeed, the subject of this course involves rich and powerful intellectual machinery for solving tremendously important technological problems. Most of you will have to work very hard to become competent in Transport Phenomena ... as has everyone that I know. One of the common myths of students is that "they are not talented enough or smart enough" to learn this subject well (in addition to other myths) I believe that each of you are plenty bright enough to master this subject beyond your wildest expectations if you have the desire and will to work at it very hard.

This is an extremely important course in the Chemical Engineering Curriculum. Many of my colleagues that are practicing engineers cite a course such as this one as a key element in the development of their own engineering skills. Many of you will learn a very new way to think about describing physical phenomena one of the important components of the course is to attain the ability to "derive," based on engineering balances, a differential equation that describes a situation of interest and to solve this equation for use in design or understanding of engineering equipment.

SOFT SKILLS READINGS

As mentioned in the section regarding “Homework,” part of the homework assignment will involve a short reading (5 pages or less) on “Soft Skills.” These readings involve other skills that you will find important and useful on the job in the future. There will be a few questions on each HW recitation quiz (10%) involving the soft skills readings.

RECITATION MEETINGS:

In addition to administering the recitation quizzes and mid-term exams during these meetings, on days following a recitation quiz there will likely be an (optional) lecture over some of the materials that I have asked you to read from the book (or my notes). I have noticed in the past that sometimes these topics are not so straightforward so I have prepared notes and will lecture about these notes for those that would like to attend.

EXAMS:

Midterm and final exams will be *very* challenging and you will need to study extensively to perform well. These exams will be closed book and closed notes. **You will be allowed one "formula card" made up of a 3 in. x 5 in. card** with anything that you like written on it (both sides may be used). **For the final exam you can use an 8.5 in. x 11 in. piece of paper for your formula card.** Calculators will not generally be allowed during these exams. **Exams cannot be made up ...** if a Midterm Exam must be missed due to illness or family crisis then the Final Exam score will be substituted for the missed exam. This can only be done once and students that must miss more than one Midterm Exam should see the Deans Office (ECJ 2.200) regarding dropping the course.

If you have questions about the manner in which your exam was graded then please write a note describing the discrepancy, attach the note to your exam and return the note and exam to me and I will review this together with the grader. This procedure must be followed within one class period of the time the graded exams are initially returned otherwise we will not re-grade your test. In addition, we will review the entire exam to look for any other discrepancies (positive or negative).

EVALUATION:

Recitation HW Quizzes*	15%	- Drop lowest single rec. quiz grade.*
Lecture Quizzes&	10%	- 3 lowest quizzes dropped.&
Midterm Exams#	18.33%	- each (3 exams for a total of 55%)#
Class Impression^	+3 or -10%	- Professionalism/participation^
Final Exam	<u>20%</u>	
	103%	- Possible Final Percentage

*If you must miss a recitation quiz then that will be the quiz grade that you will drop until you reach one. Additional missed recitation quizzes beyond one will be scored zero.

& If you must miss a lecture quiz then that will be a quiz grade that you will drop until you reach three. Additional missed lecture quizzes beyond four will be scored zero.

#You may substitute the grade you earn on the Final Exam for the lowest score you earn on a Midterm Exam (provided all three Midterm Exams are attempted and your Final Exam grade is higher than at least one of your Midterm Exam grades).

^Make me remember you for good reasons.

Final grades in the class will be determined as follows:

103-90%	A
<90-80%	B
<80-70%	C
<70-60%	D
<60%	F

DO NOT USE THE AVERAGE CALCULATED BY CANVAS TO DETERMINE YOUR STANDING IN THE CLASS – THIS IS CALCULATED AUTOMATICALLY IN SOME ARBITRARY MANNER

If you fail to score nearly perfectly on the first two recitation quizzes, I strongly encourage you to drop the class. **Poor performance on these initial two quizzes by previous students has almost always led to a final course grade of D or F.** The secret to performing well in this course is to study actively and work very hard early in the semester.

The University of Texas at Austin provides, upon request, appropriate academic adjustments for qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-4241 TDD or the College of Engineering Director of Students with Disabilities at 471-4382. Finally, scholastic dishonesty will not be tolerated and incidents of dishonesty will be reported to the Dean of Student Affairs.

ChE 319 GRADE DISTRIBUTION FALL 2022

Enrollment First Class Day – 50
Enrollment Last Class Day – 36
A – 8
A- – 2
B – 8
C+ – 4
C – 8
D – 4
F – 2

COURSE CONTENT:

Heat Transfer: Conduction - 1-Dim., Transient, Rectangular Coord., Cylindrical Coord.
Convection - Laminar Flow, Turbulent Flow, Detailed analysis and balances, Use of Heat Transfer Correlation's, Development of Heat Transfer Correlation's

Fluid Mech.: Macroscopic balances, microscopic balances.
Laminar and turbulent flow.

Mass Transfer: Ficks law, simple diffusion problems, mass transfer correlation's.

IMPORTANT DATES: (<https://registrar.utexas.edu/calendars/25-26>)

Jan. 12 – Mon. – First class period for ChE 319

Jan. 12 – Mon. – Mathematics Review Session CPE 2.216 5-6:30pm

Jan. 13 – Mon. – Mathematics Review Session CPE 2.216 5-6:30pm

Jan. 14 – Wed. – Second class period and first recitation period for ChE 319 PHOTO's & Lecture.

Jan. 19 – Mon. – Martin Luther King Holiday

Jan. 20 – Tues. – Last day to add a class; after this date changes in registration may require the Chairperson's and Deans approval.

Jan. 21 – Wed. – 4th class period for ChE 319 – **First Lecture Quiz [most lecture periods will have at least one lecture quiz from this date forward].** Recitation Quiz over Hw#1 and Syllabus.

Jan. 28 – Wed. – Recitation Quiz over Hw#2. Last day to drop a class without permission (health or serious personal problems). This problem cannot have existed on or before Jan. 28.

Feb. 4 – Wed. – Recitation Quiz over Hw#3

Feb. 11 – Wed. – Recitation Quiz over Hw#4

Feb. 18 – Wed. – **Midterm Exam #1** over lecture topics 1-6 and HW's 2-4. During Recitation 4-6pm.

Feb. 25 – Wed. – Recitation Quiz over Hw#5

Mar. 4 – Wed. – Recitation Quiz over Hw#6

Mar. 11 – Wed. – Recitation Quiz over Hw#7

Mar. 16-21 – **Spring Break**

Mar. 25 – Wed. – **Midterm Exam #2** over lecture topics 7-13 and HW's 5-7. During Recitation 4-6pm.

Apr. 1 – Wed. – Recitation Quiz over Hw#8

Apr. 8 – Wed. – Recitation Quiz over Hw#9

Apr. 15 – Wed. – Recitation Quiz over Hw#10 - Last day an undergraduate student may: Q-drop a class; withdraw; change a class to pass/fail.

Apr. 22 – Wed. – **Midterm Exam #3** over lecture topics 15-23 and HW's 8-10. During Recitation 4-6pm.

Apr. 27 – Mon. – Last ChE 319 Class Day - Last Class Day for All Classes.

Apr. 28-29 – Tues.-Wed. – Final Exam Study Days

May 1 – (Friday) 10-30 am – 12:30 pm – **ChE 319 Final Exam** over lectures 1-28 and HW's 2-10.

POTENTIALLY USEFUL BOOKS:

Engineering Library

J. P. Holman, *Heat Transfer*, 5th Ed., QC 320, H64, 1981.

J. P. Holman, *Heat Transfer*, 6th Ed., QC 320, H64, 1986.

S. Whitaker, *Introduction to Fluid Mechanics*, QA 911, W38.

M. N. Ozisik, *Heat Transfer: A Basic Approach*, TJ 260, O96, 1984.

R. B. Bird, et al., *Transport Phenomena*, QA 929, B5, 1960.

H.S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, QC 321, C28, 1959.

W. M. Kays, *Convective Heat and Mass Transfer*, 2nd Ed., QC 327, K37, 1980.

W. M. Kays and M. E. Crawford, *Convective Heat and Mass Transfer*, 3rd Ed., QC 327, K37, 1993.

E. Kreyszig, *Advanced Engineering Mathematics*, 7th Ed., QA 401, K7, 1992.

W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations and Boundary Value Problems*, 5th Ed., QA 371, B773, 1992.

J. R. Welty, C. E. Wicks and R. E. Wilson, *Fundamentals of Momentum, Heat and Mass Transfer*, 3rd Ed., TA 357, W45, 1984.

J. R. Welty, C. E. Wicks and R. E. Wilson, *Fundamentals of Momentum, Heat and Mass Transfer*, 2nd Ed., TA 357, W45, 1976.